V O L T S L O T

VoltSlot

EV Charging Station Management and Slot Reservation System

API Reference

Student name	[Enter student name]
USN	[Enter USN]
Department	[Enter department]
College	[Enter college name]
Project guide	[Enter guide name]
Academic year	[Enter academic year]

Prepared from the VoltSlot source code · 2026-10-14

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API Overview

Concise reference for the routes mounted by the VoltSlot Express application.

Base prefix: `/api`. JSON is used for request/response bodies. JWT-protected requests send `Authorization: Bearer <token>`. Backend roles are `customer` and `admin`.

Authentication and Account

Method / path	Access	Payload / result
GET /health	Public	Returns `{ api: "ok", database: "connected|disconnected" }`; status 200 or 503.
POST /auth/register	Public	`{ name, email, password }`; optional role must equal customer. Returns 201 with message, token, public user.
POST /auth/login	Public	`{ email, password }`; returns 200 with token and public user; invalid credentials 401.
GET /auth/me	JWT	Returns current public user; invalid/missing account token 401.
PATCH /auth/me	JWT	`{ name?, email? }`; only these fields; returns updated public user.
GET /user/dashboard	customer	Returns own booking summary, upcoming bookings and recent bookings.
GET /admin/dashboard	admin	Returns station/slot counts, booking statuses, recent bookings and top stations.

Stations and Slots

Method / path	Access	Payload / query / behavior
GET /stations	Public	Optional city, connectorType, status, availability, search, longitude, latitude, maxDistance filters; stations include availableSlots count.
GET /stations/:id	Public	Station detail. Invalid ID 400; missing station 404.
POST /stations	admin	Create station using editable fields: name, description, address, city, location, connectorTypes, pricePerKwh, openingHours, status. 201.
PUT /stations/:id	admin	Update at least one station editable field.
DELETE /stations/:id	admin	Delete only with no dependent slots/bookings; conflict 409 otherwise.
GET /stations/:stationId/slots	Public	List slots sorted by startTime.

Method / path	Access	Payload / query / behavior
POST /stations/:stationId/slots	admin	`{ startTime, endTime }` as timestamp strings; status defaults to available.
PUT /stations/:stationId/slots/:slotId	admin	Accept schedule fields and/or status available/unavailable. Booked/dependent records prevent changes.
DELETE /stations/:stationId/slots/:slotId	admin	Delete unreferenced slot only; dependent booking history conflicts.

Station body details

`location: { type: "Point", coordinates: [longitude, latitude] }`; coordinate bounds are longitude −180..180 and latitude −90..90. Connectors are CCS, CHAdeMO, Type 1, Type 2, GB/T, Tesla. Opening hours use HH:mm. Operational status is active/inactive/maintenance.

Bookings

Method / path	Access	Behavior / payload
POST /bookings	customer JWT	`{ slotId, energyKwh }`; requires valid slot ID and positive finite energy. Claims a future available slot, checks active station, creates confirmed booking with calculated price. Admin is rejected with 403; slot conflict 409.
GET /bookings/my	JWT	List records where customer reference matches token subject.
GET /bookings	admin	List all bookings.
GET /bookings/:bookingId	Owner or admin	Read one booking; other customer receives 403.
PATCH /bookings/:bookingId/cancel	Owner or admin	Confirmed booking before start only; admin can cancel another user. Cancellation releases slot.
PATCH /bookings/:bookingId/complete	admin	Confirmed booking only, after slot end.

> **Authorization:** `POST /bookings` requires a customer JWT. Separately, distinct slot records with overlapping time ranges are not globally excluded by the current data model/indexes.

Authorization and Errors

Status	Meaning in current API
400	Invalid body/query/ID or model validation failure.
401	Missing/invalid/expired JWT, deleted token account, or authentication configuration error.
403	Insufficient role or access to a booking owned by another customer.

Status	Meaning in current API
404	Requested resource/account not found.
409	Duplicate schedule, booking race/conflict, dependent record, or invalid state transition.
413	JSON body exceeds configured 100 KB limit.
429	Login or registration request limit exceeded.
500 / 503	Unexpected API failure / health check indicates database disconnected.

Generated from handlers under `backend/routes/` and app middleware. No API examples expose live records or credentials.